Comments on: “Navigating Change: Google Maps, Real-Time Information, and Transit Ridership: Comment” by R. Jerch, A. La Nauze, X. Peng

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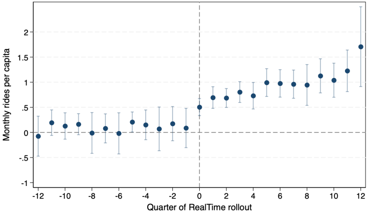

- Unit is a transit district month $N = 128$.
- x axis is quarters from rollout of real time google maps transit features (2011-19).
- y axis riders per month per capita from NTD (BJS estimator $\implies$ no omitted month).
- Treatment effect is about 1 ride month/pp.
- Base is about 7 rides pp/month, so this is a 14% increase.
- Federal budget for transit is about 50b a year.

Could this possibly be true?

At best, google maps cuts wait time to zero. Can we guess how important this would be?

- Value of wait time per minute 0.87(bus off-peak USD2002) from Parry and Small AER 2009.
- Elasticity of passenger demand wrt fare 0.80 (bus off-peak)
- Suppose (1) google maps decreases generalized cost of trip by 20% (2) fare and generalized cost elasticity are the same.
- $0.2 \times 0.8 = 0.16 \approx 0.14$. In the ballpark.
- Do this calculation more carefully.

Subways vs. Buses

- New York is about 70% of all subway rides. Top six subway cities are above 90%. Top six transit cities account for about 40% of all bus rides.
- Weighting by riders vs transit districts changes this from a paper about lots of little bus districts to a case study of NY (and maybe 5 other places).
- I think you are better set up to describe lots of little bus districts. Focus on this, drop big subway cities from main analysis and treat them in a little section at the end?
- Related #1. Except for NYC, I think buses carry at least as many riders as subways almost everywhere. Is figure A.III really just reporting on NY?
- Related #2. Are results for complicated/real time subsets both driven by NYC and other large districts? (more below).

Econometrics I

Recall the logic of the DiD/BJS estimator

- Potential outcomes for treated and untreated states,

$$Y_{it}(0), Y_{it}(1), t = 1, 2.$$

- Want

$$ATT = E(Y_{i2}(1) - Y_{i2}(0)|D_i = 1).$$

  ... but $Y_{i2}(0)|D_i = 1$ is never observed.
- BJS impute $Y_{i2}(0)$ from untreated units, usually

$$\widehat{Y_{i2}}(0) = \widehat{\alpha}_i + \widehat{\beta}_2 + \widehat{\gamma} x_{i2}.$$

- I can impute $\widehat{Y_{i2}}(0)$ from the price of fish alone and, as long as I maintain conditional parallel trends, the logic of the estimator goes through. However, (1) I am shifting from a non-parametric to a parametric imputation function, (2) the model of treated unit counterfactual untreated state is less credible a priori.

Econometrics II

Here is the paper's imputation function,

$$\widehat{Y_{it}}(0) = \eta_i + [\gamma_{dy} + \tau_{ds} + \rho_{iy}]$$
$$\eta_i = \text{City}, \ \gamma_{dy} = \text{Region Year}, \tau_{ds} = \text{Region Season},$$
$$\rho_{iy} = \text{City Year controls}, \ t, l, s = \text{Month, Quarter, Year}$$

Here is the conventional version

$$\widehat{Y_{it}}(0) = \eta_i + \beta_t$$

- The LHS has variation at the month level. Right is just quarters. Why move away from non-parametric imputation?
- With BJS, calculate ATT one month and city at a time, and aggregate $\Longrightarrow$ the whole paper needs only one regression. Get subsample results by averaging a subset of unit ATT's.

Econometrics III

- I think control set is not-yet-treated. Say so and maybe expand this to more districts?
- My guess is that this won't matter qualitatively.

Summary

- Estimate a huge effect of an information treatment.
- Looks like it could be consistent with other things we know, but make sure.
- Econometrics are probably OK, but make sure.
- I think this is a paper about lots of little bus districts. This is OK, buses are the most important transit mode almost everywhere. But try to be clearer about this.